Abdullah Ibne Tayeb Tamur

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Portfolio | GitHub | Google Scholar | ORCID | LinkedIn

RESEARCH PROFILE

Research Interests: Medical AI, Explainable AI (XAI), Multimodal Clinical AI, Medical Image Analysis, Federated Learning, and Clinical Decision Support Systems.
First author on four IEEE-indexed publications, including CHD-Net (99.15% accuracy, Best Presenter Award, IEEE QPAIN 2026) and HemX (13,823 patients, TRIPOD+AI validated, under review at *Results in Engineering*, IF 9.4, Q1). Research focuses on building clinically deployable, interpretable deep learning systems that combine predictive performance, rigorous validation, and practical deployment for real-world healthcare settings.

EDUCATION

International Islamic University Chittagong (IIUC) — *Bangladesh*
Bachelor of Science in Computer Science and Engineering — Jan 2022 – Jun 2026
CGPA: 3.40 / 4.00 *Last 4 semesters GPA: 3.69 / 4.00*
Relevant Coursework: Machine Learning, Deep Learning, Digital Image Processing, Computer Vision, Neural Networks & Fuzzy Systems, Pattern Recognition & Image Processing
Thesis: *Dual-Stream FreqViT: FFT and Vision Transformer Fusion for Robust Plant Disease Detection* (Defense pending, June 2026)

RESEARCH EXPERIENCE

Elite Research Lab LLC — *Remote*
Research Intern — Jun 2026 – Present
Conducting applied research on Explainable AI, clinical diagnostic systems, Retrieval-Augmented Generation, and Dual-Mode LLM frameworks under formal NDA. Developing interpretable machine learning pipelines for healthcare applications with attention to clinical usability, privacy compliance, and deployment constraints.

Computational Intelligence and Society (CIS) Lab, IIUC — *Chattogram, Bangladesh*
Research Mentor — Jan 2024 – Present
Mentoring undergraduate researchers through the end-to-end publication pipeline: literature review, experimental design, model development, and manuscript preparation. Guided multiple project teams toward IEEE conference submissions; supported implementation in PyTorch, computer vision architectures, attention mechanisms, and XAI tools.

PUBLICATIONS

Under Review

[1] **Tamur, A. I. T.** et al. (**First Author**), *HemX: Hierarchical XAI for Hematological Triage.* Under review at **Results in Engineering** (Elsevier, IF 9.4, Q1).
Explainable hematological triage framework evaluated on 13,823 patient records from three countries. Integrates global SHAP, patient-level LIME, and a novel HHI biomarker for cross-site normalisation. Validated under TRIPOD+AI guidelines with patient-level stratified 5-fold cross-validation.

[2] **Tamur, A. I. T.** et al. (**Second Author**), *Fed-NeuroTriage: Federated Distillation Architecture for Psychiatric NLP.* Under review at **JSCDM**.
Privacy-aware federated learning framework for psychiatric NLP triage (Bipolar vs. MDD) using MIMIC-IV discharge summaries. Addresses 250:1 class imbalance via knowledge distillation and inverse-frequency weighting.

[3] **Tamur, A. I. T.** et al. (**First Author**), *CMA-Fusion: Cross-Modal Attention Fusion for Alzheimer's Disease Detection via Multimodal Neuroimaging.* Under review at **ACM ICCA 2026**.
Multimodal framework combining structural MRI and functional PET data via cross-modal attention fusion for early-stage Alzheimer's disease detection.

[4] *DiaXAI-Stack: An Explainable AI Framework for Diabetes Prediction.* Under review. **Fifth Author & Research Mentor**.
Mentored a junior CIS Lab team through dataset preprocessing, SHAP/LIME integration for clinical transparency, and end-to-end manuscript development.

Peer-Reviewed Publications

[5] **Tamur, A. I. T.** et al. (**First Author**), *CHD-Net: An Explainable Deep Learning Framework for Pediatric Congenital Heart Disease Classification.* **IEEE QPAIN 2026. Best Presenter Award.**
Explainable ensemble framework for pediatric CHD classification from echocardiography. Achieved 99.15% accuracy and 99.99% AUC on 4,140 echocardiography images (derived from 828 source scans) using patient-level stratified 5-fold cross-validation with zero frame leakage. SHAP GradientExplainer applied for anatomical localisation of IAS, IVS, and ductus arteriosus regions.
DOI: 10.1109/QPAIN69676.2026.11545695

[6] **Tamur, A. I. T.** et al. (**First Author**), *Dual-Stream FreqViT: FFT and Vision Transformer Fusion for Robust Plant Disease Detection.* **IEEE QPAIN 2026.**
Dual-stream architecture combining a spatial CNN branch, 2D FFT spectral branch, and Transformer encoder fusion. Achieved 97.75% accuracy under challenging field-lighting conditions.
DOI: 10.1109/QPAIN69676.2026.11546204

[7] **Tamur, A. I. T.** et al. (**Third Author**), *FUSE SENet: Fusion with SE Attention Network for Robust Skin Lesion Classification.* **IEEE ICCIT 2025.**
Fusion-based feature learning with squeeze-and-excitation (SE) attention for robust skin lesion classification.
DOI: 10.1109/ICCIT68739.2025.11491158

AWARDS & HONORS

Best Presenter Award — IEEE QPAIN Conference 2026, awarded for the presentation of the *CHD-Net* clinical AI framework.

AgentX AI Prompting Competition — Ranked **6th among 700+ university participants** (IIUC-level) and placed among the **top national performers** from 50,000+ competitors nationally (Bangladesh).

DEPLOYMENTS & ENGINEERING PROJECTS

HemaTriage [Live]: Clinical web application based on the HemX architecture for CPU-only hematological triage inference in resource-constrained healthcare settings.

Customer Support Agent [GitHub]: Production-grade LangGraph + Groq (Llama 3.3 70B) agent with triage classification, six domain-specific tools (customer lookup, invoice retrieval, refund management), input/output guardrails, human-in-the-loop approval, short-term memory, and full LangSmith observability. SQLite persistence across sessions.

SpectraLeaf Analytics [Live]: Cloud-deployed plant disease detection application combining spatial CNN features with FFT-based spectral representations for robust field-condition inference.

Wolf Scholar [GitHub]: Multimodal RAG-based research assistant using LangChain, FAISS, and Groq for multi-document analysis and medical vision diagnostic workflows.

Autonomous FPV Drone: Arduino and GPS-based autonomous navigation system with 1.5 km operational range and real-time video transmission.

LEADERSHIP & SERVICE

Assistant Social Welfare Secretary, IIUC Computer Club — Organised mental health awareness programmes and coordinated peer academic support initiatives for the university computing community.

Volunteer — 2024 Flood Relief Operations — Collaborated with As-Sadaqa and the Bangladesh Red Crescent Society to prepare and distribute emergency relief packages for flood-affected communities across Chittagong Division.

Home Tutor *2020 – Present*
Teaching secondary and higher-secondary students in Biology, English, ICT, and Chemistry; providing personalised academic support and exam preparation guidance for five years alongside undergraduate studies and research.

Other: **Managing Director**, City I Restaurant, Chattogram (Jan 2023 – Present) — Operational and strategic leadership of a 45-member team alongside full-time academic and research activity.

TECHNICAL SKILLS & METHODOLOGIES

Research Methods: Explainable AI, SHAP, LIME, Federated Learning, LangGraph Agents, Conformal Prediction, Uncertainty Quantification, Patient-Level Stratified K-fold CV, TRIPOD+AI, PROBAST+AI

Languages: Python, C++, SQL, HTML/CSS, JavaScript

Frameworks & Tools: PyTorch, TensorFlow, Keras, OpenCV, LangChain, LangGraph, Flask, FAISS, Git, LaTeX, Jupyter Notebook